

Joshua Sternadel

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Systems Architect & Principal Infrastructure Engineer

A high-capacity Systems Architect and U.S. Patent author leveraging advanced spatial thinking and first-principles engineering to resolve complex structural debt. A self-directed technical anchor with 16 years of tech experience, including 13 years driving radical automation, distributed systems design, and continuous DevSecOps compliance across massive Fortune 500 telecom environments. Proven track record of securing mission-critical stability through major corporate mergers, high-stakes enterprise cloud migrations, and complex algorithmic problem-solving.

Core Competencies

Team Leadership | Systems Architecture | Solutions Engineering | Efficiency and Productivity Specialist

Key Achievements

- **Strategic Retention & Continuity:** Awarded a significant retention bonus during the Sprint/T-Mobile merger, designated as mission-critical personnel to ensure technical stability and architectural integration during the corporate transition.
- **U.S. Patent Author (#10,366,069):** Co-authored a patent for database management systems, specifically engineering a natural sorting algorithm in SQL to optimize data retrieval and user experience.
- **Infrastructure Transformation:** Orchestrated a full-scale migration from legacy physical servers to a redundant VMware-based virtual infrastructure, reducing enterprise downtime by over 90% and cutting operational payroll costs by 18%.
- **Algorithmic Solutions:** Engineered a first-principles solution to resolve GPS signal noise and “teleportation” errors for intermodal logistics. By deciphering raw NMEA 0183 sentences and applying spherical geometry with rate-of-acceleration modeling, successfully stabilized coordinate paths and eliminated tracking jitter within high-interference rail yards.

Professional Experience

T-Mobile | Solutions Architect | Senior Systems Engineer | 2021 to 2026

Embedded within Tools Team to assist with architectural duties and build APIs that connect proprietary

systems to Atlassian and other vendors' products. Transitioned to DevOps role as the only true developer on the SysAdmin team.

- **Enterprise Cloud Migration:** Architected and executed the high-integrity migration of a decade's worth of metadata, complex workflows, and mission-critical certification data from Jira Data Center to Jira Cloud; successfully preserved zero-loss continuity across 300+ proprietary device projects supporting 3,000+ active enterprise users.
- **Ecosystem Integration & APIs:** Authored a specialized suite of Node.js libraries and Node-RED nodes to automate full-lifecycle project provisioning via Atlassian APIs, creating a "self-service" ecosystem for developers.
- **Infrastructure as Code (IaC):** Bridged the gap between Software Engineering and Systems Administration by implementing Infrastructure as Code (IaC) via Ansible, successfully reducing manual operational toil and human error.
- **DevSecOps & Automation:** Engineered automated vulnerability remediation workflows that drove lab data center security debt from 100+ monthly criticals to zero, achieving a state of continuous compliance.
- **Value Stream Optimization:** Optimized the VM lifecycle by automating the end-to-end onboarding process, significantly accelerating the "Lead Time to Deployment" for internal engineering teams.
- **Full-Lifecycle Application Engineering (Angular & Serverless):**
 - Authored and launched a high-visibility 5G Speed Calculator application utilizing Angular and a modern serverless deployment pipeline to ensure instant scalability and zero infrastructure overhead.
 - Designed and engineered a custom Device Pre-Load Feature Enablement Tracker, providing corporate stakeholders with real-time visibility into baseline software states across mobile devices.
 - Bridged the gap between UI/UX design concepts and technical feasibility, establishing strict component styling and layout standards to ensure frontend performance and visual consistency.

Sprint | Principal Engineer | Application Developer III | 2017 to 2021

Served as Core Technical Advisor and Architectural Gatekeeper to leadership, maintaining absolute technical veto power and engineering oversight for all departmental systems and cross-functional integrations.

- **Architectural Leadership:** Directed the development of a "First Principles" replacement for legacy SKU management, delivering a high-performance solution in 6 months that outpaced a decade of failed initiatives by multiple cross-functional teams.
- **Distributed Systems:** Designed and deployed "Porcupine," a custom event-driven message bus architected on top of RabbitMQ; seamlessly synchronized live Jira workflow states with downstream application environments, establishing a single source of truth and eliminating cross-system data lag.

- **Framework Engineering:** Engineered a custom, data-agnostic PHP framework utilizing reflection and annotations to automate the CRUD lifecycle across disparate schemas (SQL, NoSQL, JSON, XML).
- **Data Abstraction & Hydration:** Developed a model-driven hydration engine that abstracted away storage-layer complexity, allowing for seamless state transitions between relational and non-relational data sources.
- **Scale & Adoption:** Prototyped and deployed the framework to internal high-value T-Mobile applications, proving a 10x reduction in “boilerplate” code for new service onboarding.
- **Data Architecture & Engineering Prototyping:**
 - Architected and validated a standalone Dremio data lakehouse POC designed to federate data layers across distributed MS SQL, MySQL, and MongoDB environments.
 - Proved the viability of zero-ETL query federation across relational and NoSQL silos, establishing a blueprint for unified data access.
- **Observability & Infrastructure Telemetry:**
 - Built an integrated observability platform using the ELK Stack, Prometheus, and Grafana across a 45-VM development environment.
 - Designed end-to-end infrastructure telemetry and automated alert routing hooks via Rocket.Chat to demonstrate proactive incident response frameworks.
 - Engineered a custom heartbeat monitoring framework embedded within distributed .NET Core applications; piped real-time telemetry directly into Grafana to ensure proactive incident response and eliminate production blind spots.
- **Financial Stewardship & Capital Forecasting:**
 - Managed and optimized the departmental infrastructure and licensing budget as it scaled from an annual spend of \$300K to \$1.3M.
 - Formulated highly accurate 12-month financial and capacity projections to align capital expenditures with rapid system expansion.

Sprint | Product Development Engineer II | 2013 to 2017

Transitioned from device testing into application development through hybrid role bridging both disciplines.

- **Full-Stack Application Architecture:** Engineered a specialized visual prototyping platform featuring a drag-and-drop grid system to simulate and design Out-of-Box Experiences (OOBE); integrated background profiling engines to capture real-time CPU, RAM, storage, and battery metrics alongside automated Play Store data scraping.
- **Database Engineering & Tooling:** Authored the foundational codebase for the Device Development and Certification requirements and test-plan database, laying the groundwork for scalable documentation and testing.

- **Cross-Functional Advisory:** Advised the Retail Department in weekly technical reviews, clarifying bugs, known issues, and system behaviors so teams could plan and prioritize work with greater accuracy.
- **Release Readiness & Prioritization:** Planned and prioritized weekly device onboarding for the Slate platform, improving visibility and accelerating readiness across teams.

EXPERIS (Contractor at Sprint) | Technical Support Rep II | 2012 to 2013

Developed, tested, and troubleshoot XML scripts for the SLATE platform, which retail stores use for automated device testing.

- Wrote XML scripts for Slate automation testing of mobile devices.
- Worked with OEM partners, troubleshooting issues with the Slate Device Testing platform.

MO-KAN Container Services | System Administrator | 2011 to 2012

- Managed all technology operations across 3 onsite locations, keeping systems stable and responsive for daily business needs.
- Planned and executed a full network modernization, replacing aging G3 servers with a VMware-based virtual infrastructure and scalable iSCSI SAN.
- Cut IT downtime by more than 90% through a shift to virtualized, redundant systems.
- Reduced IT payroll costs by nearly 18% by upgrading infrastructure and implementing automated, failover-ready environments.

Additional Relevant Experience

Trucking Central | Software Consultant (Contract)

Planned, developed, and tested Android-based GPS solutions for the intermodal supply chain industry.

Innovative Solutions | Mobile Solutions Technician

Troubleshoot and fixed Nextel devices for first responders and construction businesses, listened to client needs, and leveraged emerging technologies to find solutions for those needs.

Community Impact & Leadership

Seek and Serve | Technical Advisory Board Member | 2026 to Present

- **Technical Guidance:** Provide high-level architectural oversight and logistical strategy to the parent 501(c)(3) organization.

33meals.org (Subsidiary) | Infrastructure Consultant / Volunteer | 2026 to Present

- **Organizational Launch:** Spearheaded the launch and incorporation of 33meals.org as a dedicated subsidiary to address food insecurity through scalable, data-driven, distribution models.
- **Operational Infrastructure:** Managed the full technical and administrative setup, including domain registration, digital architecture, and regulatory filings to ensure high-integrity non-profit operations.

- **Strategic Development:** Designed a streamlined logistics model to optimize charitable food delivery, applying enterprise efficiency principles to humanitarian service.

Education

Associate of Arts (AA), Graphic Design | Johnson County Community College

Technical Environment Matrix

- **Architectural Disciplines:** Systems Architecture, Distributed Systems, Microservices, API Gateway Design, Platform Abstractions, SRE, DevSecOps
- **Frontend Web Frameworks:** Angular (Pre-1.0 to v15), Vue.js 2, TypeScript, JavaScript (ES6+), Client-Side Architecture
- **Backend Environments & Languages:** Node.js Runtime, .NET Core (2.x), C#, Native Modules, Zero-Dependency Design
- **Infrastructure & Automation:** Ansible, DevOps, CI/CD, Infrastructure as Code (IaC), Automation, Site Reliability, Serverless, Red Hat Linux, VMware
- **Data & Ecosystems:** Dremio (Lakehouse Federation), ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Rocket.Chat, PostgreSQL, SQL, MongoDB, Atlassian SDK, Jira API (US Patent #10,366,069)